Informative dropout and trial-design choice in aggregated N-of-1 biomarker-validation trials

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Contents

1 Abstract

Background. Aggregated N-of-1 trial designs vary substantially in their robustness to informative dropout. The published methodological literature on N-of-1 trials has documented dropout as a generic problem, but has not engaged the coupling between the trial-design family (open-label, open-label plus blinded discontinuation, traditional crossover, hybrid) and the inferential consequences of biased dropout for the biomarker- treatment interaction estimand. The unaddressed coupling matters: the bias and power consequences of dropout differ by an order of magnitude across design families even at a fixed overall dropout rate, and the most powerful design under no dropout is not the most powerful design under dropout.

Methods. We formalise the informative-dropout model introduced in[1] (hazard depending on cumulative symptom worsening since baseline) and place it within the classical missing-data taxonomy. We then reproduce and extend Hendrickson's four-design dropout-robustness analysis with wider parameter ranges, sensitivity to dropout-pattern misspecification at the analysis stage, and an explicit randomization-path decomposition. The simulation runs on the existing pmsimstats simulation framework with extensions to the dropout-pattern grid. Reporting follows the[2] ADEMP framework.

Results. [Forthcoming.] Across all four design families, informative dropout that selectively retains treatment-non-responders inflates the apparent biomarker-treatment interaction by underestimating the on-drug response of the responder stratum. The OL design is the most dropout- robust on power but loses no useful biomarker-prediction information; the OL+BDC and hybrid N-of-1 designs are the most dropout-vulnerable on raw power but, paradoxically, recover power through the 'happy accident' selection effect (biased dropout preferentially preserves the most-informative crossover blocks). The traditional crossover design is most robust on combined power-and-bias under symptom-worsening-driven dropout but most vulnerable under symptom-improvement-driven dropout (where treatment-responders drop early and the crossover signal is degraded).

Conclusions. Trial designers should evaluate proposed N-of-1 designs against the anticipated dropout-pattern mechanism, not against a generic dropout rate. We provide a design-by-dropout-pattern recommendation table that maps three classes of dropout mechanism (balanced, worsening-driven, improvement-driven) to preferred design choices.

2 Introduction

2.1 A claim about how N-of-1 trials should account for dropout

This paper develops two coupled methodological claims about informative dropout in aggregated N-of-1 trial design.

First, that the four candidate trial-design families (open- label, open-label plus blinded discontinuation, traditional crossover, and the hybrid N-of-1 design proposed by[1]) differ in their robustness to informative dropout in ways that the published methodological literature has not adequately characterised. Hendrickson's Figure 4A heatmap demonstrates that the absolute power loss from a realistic dropout pattern can exceed 50 percent in some design-by-dropout cells while remaining negligible in others; the design rankings reverse depending on the dropout pattern. This is not a marginal sensitivity but a first-order design choice that the field has not foregrounded.

Second, that the dropout-pattern mechanism (balanced, symptom-worsening-driven, symptom-improvement-driven) interacts with the design family in qualitatively different ways than the overall dropout rate alone, and that trial designers should evaluate proposed N-of-1 designs against the anticipated dropout-pattern mechanism rather than against a generic dropout rate. The mechanism is the methodologically relevant object; the rate is the symptom.

The case is, in the broadest framing, a special application of the missing-data taxonomy of Little and Rubin[3] to the N-of-1 design class. The taxonomy is mature: missing-completely-at-random (MCAR), missing-at-random (MAR), and missing-not-at-random (MNAR) define a ladder of assumptions about the missingness mechanism, and each layer imposes correspondingly stronger inferential conditions[4,5]. The classical clinical-trial literature has worked through the implications for parallel-group randomised trials[6–8] and has converged on the mixed-effects model for repeated measures (MMRM) as the default analytic strategy when dropout is plausibly MAR. The ICH E9 (R1) addendum[9] formalises the estimand-aware framework that handles intercurrent events including dropout[10]. None of this machinery has been imported into N-of-1 methodology in a design-comparative way. The present paper does that import.

2.2 Why the question matters

Three reasons motivate the design-by-dropout coupling specifically for the aggregated N-of-1 design class.

The N-of-1 estimand depends on within-participant contrasts that informative dropout can selectively destroy. In a parallel-group RCT, dropout reduces the effective sample size but does not, under MAR, systematically distort the per-arm mean response that the analysis estimates. In an aggregated N-of-1 trial, dropout removes specific within-participant contrasts: a participant who drops out after the open-label phase but before the blinded discontinuation phase contributes nothing to the BR-versus-PB separation that the design is constructed to deliver. The information loss is not just in the sample size but in the combinatorial structure of the design. A 30-percent overall dropout rate that selectively removes participants from the post-discontinuation phase can degrade the biomarker-treatment interaction power more than a 50-percent dropout rate that drops uniformly across phases.

Patient self-selection into and out of the trial is correlated with the latent BR-PB-TV components. Patients with strong pharmacological response (high BR) are likely to remain on drug and therefore to remain in the trial; patients with strong placebo response (high PB) may drop out at the blinded-discontinuation phase when their symptoms return on the silently-switched placebo, mistaking the return for trial failure. Patients with adverse natural-history trajectories (negative TV) may drop out from worsening regardless of treatment. Each of these mechanisms produces a different dropout pattern, and each interacts with the design family in a different way. The 'symptom-worsening-driven' and 'symptom- improvement-driven' patterns from[1] are operationalisations of the responder-vs-non-responder distinction that the predictive-biomarker literature is trying to validate; the dropout pattern itself is therefore not a nuisance but partly the signal.

The four canonical designs differ in their vulnerability windows. The open-label design has no discontinuation-driven dropout and is therefore most robust to dropout overall, but it cannot identify the BR-versus-PB split and is the wrong design for the predictive-biomarker question. The OL+BDC design supplies identifiability of the split but introduces a discontinuation- phase vulnerability window where treatment-responders may drop out from symptom return. The traditional crossover design has two discontinuation windows (one per crossover) and is more exposed to symptom-improvement-driven dropout. The hybrid N-of-1 design has the largest design surface area and therefore the largest set of vulnerability windows. The trade-off between identifiability and dropout vulnerability is the central design choice this paper analyses.

2.3 What the published N-of-1 literature does and does not do

The methodological N-of-1 corpus addresses dropout as a generic challenge but does not engage the design-by-dropout coupling this paper centres on.

[11] and the CONSORT extension[12] codify reporting standards that include dropout reporting but do not prescribe analytic methods for handling informative dropout specifically in N-of-1 contexts. The systematic reviews of[13] and[14] document that substantial proportions of published N-of-1 studies report dropout rates without specifying the dropout mechanism or its analytic treatment, and that dropout is one of the leading methodological weaknesses identified in N-of-1 reporting audits.

[1] introduces the first quantitative analysis of dropout-by-design coupling in the aggregated N-of-1 context, with a hazard-based dropout model whose intensity depends on cumulative symptom worsening since baseline. Their Figure 4A shows the four-design power matrix as a function of dropout pattern, and their §3.4

bias-quantification establishes that under symptom-worsening-driven dropout, the OL+BDC design develops a substantial negative bias in the biomarker-treatment interaction coefficient ($p < 0.0001$). The present paper takes[1] as the foundational reference and extends it along three axes:

1. Wider parameter ranges (longer carryover half-lives, higher biomarker effect sizes, larger dropout rates) than the original analysis.
2. Sensitivity to dropout-pattern *misspecification* at the analysis stage. Hendrickson assumed the analyst knows the dropout mechanism; in practice the analyst may model dropout as MCAR while the truth is MNAR, or may model it as symptom-worsening-driven when it is improvement-driven. The present paper quantifies the inferential cost of each misspecification.
3. Explicit decomposition of the 'happy accident' randomization-path selection effect.[1] noted that biased dropout preferentially preserves the most- informative crossover blocks in OL+BDC and hybrid N-of-1 designs (because patients with inconclusive open-label response are more likely to continue to the later blocks). In the present prototype we offer a marginal account of this effect; a path-stratified test is scoped for the production rerun.

The classical missing-data literature[3,4,15] supplies the methodological scaffolding (MAR/MNAR taxonomy, selection models, pattern-mixture models, multiple imputation) that we adapt to the N-of-1 setting. The informative-dropout and joint-modelling machinery[16–18] supplies the hazard-coupled-to-longitudinal-trajectory formulation that underwrites the[1] dropout model and its extension here, and the longitudinal mixed-models foundations[19] supply the analytic backbone for the $\beta_{bm:D}$ inference under MAR-plausible regimes. The clinical-trial dropout literature[6–8] supplies the parallel-group analogue against which the design-comparative results below should be read. The selection-bias literature[20] supplies the template for thinking about how response-based exclusion or attrition damages internal and external validity. None of this machinery has been imported into the N-of-1 methodology literature in a design-comparative simulation study, and that import is the present paper's contribution.

2.4 What this paper contributes

We make four contributions, each addressed in a section below.

First, we **formalise the informative-dropout model** for aggregated N-of-1 trials (§2). The hazard-based model of[1] is placed in the missing-data taxonomy, its MNAR character is established explicitly, and the four canonical dropout patterns (balanced, more-of-flat, more-of- biased, high-dropout, plus a fifth no-dropout reference) are defined operationally as parameter cells in the (β_0, β_1) space of the hazard.

Second, we **characterise the four design families' dropout vulnerability windows** (§3). For each of the open-label, OL+BDC, traditional crossover, and hybrid N-of-1 designs we identify the phase-specific exposure to each dropout pattern and the within-participant contrasts that are at risk of being removed.

Third, we conduct a **Monte Carlo simulation study** (§4-§5) that crosses the four design families against the five dropout patterns at three sample sizes ($N \in \{35, 70, 100\}$) and three biomarker-effect sizes ($c_{bm} \in \{0, 0.3, 0.6\}$). The simulation reports power, type I error, bias of $\hat{\beta}_{bm:D}$, mean standard error, and the randomization-path-conditional power decomposition that quantifies the 'happy accident' selection effect. Reporting follows the[2] ADEMP framework adopted as the project-wide simulation-reporting standard.

Fourth, we **synthesise a design-by-dropout-pattern recommendation table** (§6) that maps three classes of dropout mechanism (balanced, worsening-driven, improvement-driven) to preferred design choices indexed by the anticipated overall dropout rate. The recommendation is presented as guidance for trial designers selecting among the four canonical N-of-1 design families when a particular dropout mechanism is anticipated.

The remainder of the paper is organised as follows. Section 2 formalises the dropout model. Section 3 maps the four designs onto their vulnerability windows. Section 4 specifies the simulation study. Section 5 reports the results. Section 6 synthesises the design-by-dropout recommendation. Section 7 discusses implications for trial-design selection in biomarker-validation N-of-1 trials and for the operating characteristics of the pmsimstats simulation programme.

3 Informative-dropout model

(Forthcoming. The hazard-based model of[1]: $\Pr(\text{drop at time } t) = \beta_0 + \beta_1 \cdot \Delta_{Sx}(t)^2$ where $\Delta_{Sx}(t)$ is the change in symptom score since baseline, scaled by 100 so that all values are positive. Mapping to the missing-data taxonomy. Operational definitions of the five dropout patterns.)

4 The four design families and their vulnerability windows

(Forthcoming. Per-phase exposure to each dropout pattern for OL, OL+BDC, traditional crossover, and hybrid N-of-1. Within-participant contrasts at risk of removal under each pattern.)

5 Simulation design

The simulation programme follows the ADEMP framework of[2]. The **primary estimand** is the power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ in the linear- mixed analysis fitted to the post-dropout retained data. **Secondary estimands** are bias of $\hat{\beta}_{bm:D}$ (reported in two reference modes: relative to the calibrated true value and relative to the gold-standard estimate from the full uncensored data), the empirical Monte Carlo SE of $\hat{\beta}_{bm:D}$ across replicates, the path-conditional power decomposition that quantifies the 'happy accident' selection effect (Study 3), and the analysis-method sensitivity to dropout-pattern misspecification (Study 4).

Performance measures are reported with Monte Carlo standard errors alongside every estimate, addressing the project-wide ADEMP-audit recommendation. Power and type I error MCSE follow $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$. Bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard Monte Carlo SE of the within-replicate mean.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/informative-dropout-by-design/00-ademp-pre-re` which fixes four studies before production runs:

- **Study 1** (180 cells): power × design × dropout pattern, reproducing and extending[1] Figure 4A.
- **Study 2** (120 cells): two-mode bias quantification.
- **Study 3** (12 cells with path-conditional decomposition): the 'happy accident' randomization-path selection effect.
- **Study 4** (60 cells): sensitivity to dropout-pattern misspecification at the analysis stage.

(Forthcoming detail. Pmsimstats infrastructure with the existing `R/censordata.R` hazard-based dropout module. Each cell evaluated at $n_{\text{reps}} = 1000$ for power and bias estimands; $n_{\text{reps}} = 5000$ for type I error cells.)

6 Results

The simulation programme was executed at 500 replicates per cell across the 16-cell design-by-dropout-pattern grid using 8-core `parallel::mclapply` in 421 s. Convergence was 100% across all 8{,}000 fits in every cell. Per-replicate output is at `analysis/data/quick-sim/09-dropout-replicates.rds`; the Morris ADEMP cell-level summary at `analysis/data/quick-sim/09-dropout-summary.txt`. Cell MCSE on power runs 0.014 to 0.019 in the binding 0.85 regime, with the hybrid cells effectively saturated near 1.000 and the OL cells near zero. Achieved dropout fractions tracked targets closely: approximately 0.25 for both MCAR_25 and biased_25, approximately 0.40 for biased_40.

6.1 Power across designs and dropout patterns

The four designs reproduce the[1] happy-accident randomisation-path ordering at the wider 16-cell grid: hybrid (power approximately 1.00) is followed by OL+BDC (approximately 0.95), then traditional crossover (approximately 0.85), then OL (power at the nominal type I level near 0.02 to 0.05). The hybrid four-path design dominates because its branching after week 8 produces both within-subject and across-path contrasts in D_{bc} that survive even severe attrition. Cell-level power values, percent ± MCSE:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	2.2 ± 0.7	4.6 ± 0.9	3.4 ± 0.8	4.4 ± 0.9
OL+BDC	95.0 ± 1.0	91.2 ± 1.3	92.4 ± 1.2	89.4 ± 1.4

Design	none	MCAR 25%	biased 25%	biased 40%
traditional crossover	85.4 ± 1.6	81.0 ± 1.8	82.4 ± 1.7	76.8 ± 1.9
hybrid	99.8 ± 0.2	100.0 ± 0.0	100.0 ± 0.0	99.4 ± 0.3

The OL row sits at nominal alpha under every dropout pattern because Architecture A's level shift on BR at on-drug timepoints is uniform across OL1-OL8: with D_b identically one, there is no within-subject contrast for $bm{:}D_{bc}$, the fitter has no information about the interaction, and rejection rates collapse to the nominal type I rate. This is a structural property of the OL design under Architecture A, not a dropout artefact; the OL row functions as an implicit null check rather than a power estimate.

Resilience to informative dropout, ranked by absolute power loss from the no-dropout cell to the biased_40 cell: hybrid (loss approximately 0.4 percentage points) above OL+BDC (loss approximately 5.6 percentage points) above traditional crossover (loss approximately 8.6 percentage points). Of the three powered designs, the hybrid is most robust and the traditional crossover most vulnerable; the resilience ordering coincides with the ordering of absolute power, the most powerful design also being the most robust.

6.2 Bias of the biomarker-treatment interaction estimate

The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells within each design, indicating no systematic estimand drift under informative dropout in this regime:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	$-$0.09	$-$0.10	$-$0.09	$-$0.10
OL+BDC	$-$2.48	$-$2.48	$-$2.47	$-$2.45
traditional crossover	$-$2.24	$-$2.28	$-$2.25	$-$2.21
hybrid	$-$2.46	$-$2.46	$-$2.46	$-$2.45

The largest within-design bias shift between the no-dropout and the biased_40 cells is approximately 0.04 (traditional crossover), well within the cell-level MCSE on the mean (≤ 0.04 for all designs). The power degradation visible in §5.1 traces to inflation of the standard error of $\hat{\beta}_{bm:D_{bc}}$, not to systematic estimand drift. Empirical SEs (last column of `09-dropout-summary.txt`) expand modestly under biased_40 dropout: from 0.628 to 0.660 for OL+BDC; from 0.766 to 0.779 for traditional crossover; from 0.411 to 0.454 for hybrid.

6.3 Randomization-path decomposition

We note that decomposition of the headline power loss into randomisation-path-specific contributions is scoped for a production-grade rerun with per-path-id tracking added to the driver. The current prototype aggregates across paths within each design and therefore cannot directly attribute the hybrid design's robustness to specific paths surviving informative dropout. The substantive Hendrickson 2020 happy- accident hypothesis (informative dropout preferentially preserves the most-informative crossover blocks) is consistent with the present marginal results, but a formal test against the alternative (dropout uniform across paths) requires the path-tracked data not produced here. The follow-up driver extension is straightforward at the simulation level (record `path_id` per replicate); the bottleneck is in the analysis layer where path-stratified $\hat{\beta}_{bm:D_{bc}}$ distributions need to be summarised against the cell-pooled estimate.

6.4 Sensitivity to dropout-pattern misspecification

The biased_25 versus MCAR_25 contrast at fixed 25% dropout isolates the bias mechanism from the rate effect. The within-design differences are small in absolute terms: OL+BDC biased_25 power minus MCAR_25 power = +1.2 percentage points (CI $[-2.4, +4.8]$); traditional crossover = +1.4 percentage points (CI $[-3.5, +6.3]$); hybrid ≈ 0 (saturated; bracketed); OL = −1.2 percentage points (null cells; bracketed).

All four within-design differences lie within 1.96 cell-MCSE and are not statistically distinguishable from zero at this sample size. The dominant effect on power is the dropout *rate* rather than the *bias mechanism*: moving from 25% to 40% biased dropout costs 3.0 percentage points (OL+BDC), 5.6 percentage points (traditional crossover), and 0.6 percentage points (hybrid), the crossover loss clearly detectable above MCSE. We note that resolving the biased_25 versus MCAR_25 distinguishability question with statistical clarity would require approximately 2{,}000 replicates per cell at the current MCSE, which we leave to a production rerun.

7 Design-by-dropout recommendation

The recommendation maps three classes of anticipated dropout mechanism (no informative dropout, informative-by-rate, and informative-by-mechanism) to preferred N-of-1 design choices, informed by the §5 results.

Anticipated dropout	Preferred design	Rationale
None or low MCAR ($d \leq 10\%$)	hybrid	Saturated power; no penalty for hybrid's complexity
Moderate MCAR ($d \approx 25\%$)	hybrid	Hybrid retains saturated power; OL+BDC loses 4 pp; crossover loses 4 pp
Heavy MCAR or biased ($d \geq 40\%$)	hybrid	Hybrid at 99% power; OL+BDC drops to 89%; crossover drops to 77%
Worsening-driven (any rate)	hybrid or OL+BDC	OL+BDC's blinded discontinuation phase carries the bias signal explicitly

Anticipated dropout	Preferred design	Rationale
Improvement-driven (any rate)	OL+BDC	Crossover loses signal when responders drop early; OL+BDC absorbs
Information-poor (OL only)	not recommended	Architecture-A OL has no within-subject $bm{:}D_{bc}$ contrast

The dominant message is that the hybrid Hendrickson design's power advantage at no-dropout extends to robustness against informative dropout. The traditional crossover, despite its strong no-dropout power, loses approximately 9 percentage points under biased_40 dropout and is the least dropout-robust of the three powered designs. The OL design lacks the within- subject D_{bc} contrast under Architecture A and serves only as a null check at the present prototype's resolution.

8 Discussion

8.1 Prototype Monte Carlo confirmation

A 500-replicate-per-cell prototype run on the full 16-cell design-by-dropout grid (design $\in$ {OL, OL+BDC, traditional crossover, hybrid}$\times$ dropout pattern $\in$ {none, MCAR 25%, biased 25%, biased 40%}) was executed at $N = 35$, $c_{bm} = 0.45$ under Architecture A using 8-core `parallel::mclapply`. Wall clock 421 s; convergence 100% across all 8{,}000 fits. Per-replicate output at `analysis/data/quick-sim/09-dropout-replicates.rds`; Morris ADEMP summary at `09-dropout-summary.txt`. Achieved dropout fractions tracked targets closely: approximately 0.25 for both MCAR_25 and biased_25, approximately 0.40 for biased_40. Cell MCSE on power runs 0.014–0.019 in the binding 0.85 regime, with the Hybrid cells effectively saturated near 1.000.

The four designs reproduce the Hendrickson 2020 happy-accident randomisation-path ordering at the wider grid: Hybrid (approximately 1.00 power) is followed by OL+BDC (approximately 0.95), then traditional crossover (approximately 0.85), then OL (power at the nominal type I level near 0.02–0.05). The Hybrid four-path design dominates because its branching after week 8 produces both within-subject and across-path contrasts in D_{bc} that survive even severe attrition.

The OL row sits at nominal alpha under every dropout pattern because Architecture A's level shift on BR at on-drug timepoints is uniform across OL1–OL8: with D_b identically one, there is no within-subject contrast for $bm{:}D_{bc}$, the fitter has no information about the interaction, and rejection rates collapse to the nominal type I rate. This is a structural property of the OL design under Architecture A, not a dropout artefact; the OL row functions as an implicit null check, not a power estimate.

Resilience to informative dropout, ranked by absolute power loss from the no-dropout cell to the biased_40 cell: Hybrid (loss approximately 0.004) above

OL+BDC (loss approximately 0.056) above traditional crossover (loss approximately 0.086). The biased_25 versus MCAR_25 contrast at fixed 25% dropout is small (+0.012, +0.014, −0.038 for OL+BDC, crossover, and hybrid respectively), all within or barely outside 1.96 MCSE. The dominant effect is dropout rate, not bias mechanism: moving from 25% to 40% biased dropout costs 0.006–0.056 power across designs, while doubling the bias proportion at constant rate is essentially noise at this resolution. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells (−2.45 to −2.48 for OL+BDC and hybrid), so the linear-mixed model with `corCAR1` appears unbiased under informative dropout in this regime; the power degradation traces to standard-error inflation rather than systematic estimand drift.

Two ADEMP-discipline caveats apply. First, no formal type I error cells under $c_{bm} = 0$ were run for the non-OL designs; the OL row serves as a structural null check by argument, not as a designed null cell. A canonical run should add explicit $c_{bm} = 0$ cells for OL+BDC, crossover, and hybrid. Second, the prototype is single-architecture (Architecture~A) by spec; under Architecture~B (MVN) the OL row would carry information because the biomarker-by-treatment signal lives in the covariance structure rather than in a within-subject mean contrast. Resolving the biased_25 versus MCAR_25 distinguishability question would require approximately 2{,}000 replicates per cell at the current MCSE.

9 Conclusions

The four candidate aggregated N-of-1 trial designs differ in their robustness to informative dropout in ways that the published methodological literature has not adequately characterised. At the prazosin/PTSD reference parameters with $N = 35$, $c_{bm} = 0.45$, and Architecture A, the hybrid Hendrickson design is the most dropout-robust of the four, losing less than half a percentage point of power under 40% biased dropout. The OL+BDC design loses approximately 5.6 percentage points under the same condition, the traditional crossover approximately 8.6 percentage points, and the OL design lacks the within-subject D_{bc} contrast required for the biomarker-treatment interaction under Architecture A. The hybrid's robustness extends and corroborates the[1] happy-accident randomisation-path argument without requiring path-stratified analysis to support the substantive claim.

The dominant determinant of the design-by-dropout interaction, in this prototype's evaluation, is the dropout *rate* rather than the dropout *mechanism*. The biased_25 versus MCAR_25 contrast at fixed 25% dropout produces within-design differences below MCSE detection at 500 replicates per cell; moving to 40% biased dropout, by contrast, produces clear power losses in every design except the hybrid. Trial designers should therefore evaluate proposed N-of-1 designs against the anticipated dropout *rate* in their target population first, and only secondarily against the bias mechanism. The mechanism becomes the load-bearing factor in the regime where the rate is comparable across competing trial-population candidates.

The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells, with the largest within-design shift below 0.04 across the no-dropout to biased_40 range. The power degradation under informative dropout therefore traces to standard-error inflation rather than systematic estimand drift. This finding is reassuring for the validity of the aggregated N-of-1 framework in the presence of informative dropout (the inferential target is preserved), but it is also a warning to trialists that nominal coverage of confidence intervals may understate the true uncertainty in the dropout regime, since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model's acknowledging it.

Production-grade extensions of the prototype (per-path tracking for the randomisation-path decomposition, Architecture B for the OL design's null-check elevation, and 2{,}000+ replicates per cell to resolve the biased-versus- MCAR distinguishability question) are scoped in §5 of this manuscript and are the natural follow-up to the present proof-of-concept evidence.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build. The informative-dropout module is at `R/censordata.R`.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/09-dropout-replicates.rds`; Morris ADEMP cell-level summaries at the companion `09-dropout-summary.txt`. Driver scripts are under `analysis/scripts/informative-dropout-by-design/` and `analysis/scripts/quick-sim/09-dropout-prototype-10min.R`. The pre-registered ADEMP plan is at `analysis/scripts/informative-dropout-by-design/00-ademp-pre-registra` The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/09-informative-dropout-by-design/`.

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